

216301013

BCE-C513

SEMESTER EXAMINATION- DECEMBER 2023

CLASS: BTech SEMESTER: V

DESIGN AND ANALYSIS OF ALGORITHM

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: A and B. Attempt both sections as per the given instructions.

SECTION-A

(SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

- Question-1: List and explain the essential characteristics of algorithms. What does it mean for an algorithm to be "finite"?
- Question-2: List the types of asymptotic notations used to represent the complexity of an algorithm.
- Question-3: Why crossing sequences are essential tools for analyzing space complexity in algorithms?
- Question-4: Describe the key characteristics of problems that belong to the PSPACE complexity class. Provide an example of a real-world problem that fits into this class.
- Question-5: What is a Hamiltonian cycle? Why is finding Hamiltonian cycles an important problem in computer science and optimization?
- Question-6: Describe in brief any one sorting algorithm of your choice that is based on the divide and conquer technique.
- Question-7: Describe how Prim's algorithm is used to generate a minimum spanning tree.
- Question-8: Describe any one single-source shortest path algorithm using an example.
- Question-9: What is the relationship between P and NP classes?
- Question-10: Describe the Strassen's matrix multiplication algorithm.

SECTION-B
(LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.
(4 X 10 = 40 Marks)

- Question-1: a) Define algorithm analysis and explain its significance in the field of computer science. Why is it crucial to assess algorithm efficiency?
b) What is space complexity in algorithm analysis, and why is it important? How is space complexity represented using big O notation?
- Question-2: Explain the role of recurrence equations in algorithm analysis and design. How do they contribute to understanding algorithm performance? Provide a suitable example to illustrate your points.
- Question-3: How does parallel computing relate to the Linear Speedup Theorem, and what role does it play in reducing an algorithm's execution time?
- Question-4: Describe the traveling salesman problem and show that it is NP-complete.
- Question-5: Explain the N-Queen problem and its significance in the field of constraint satisfaction problems. Provide an example of how it can be applied in a real-world context.
- Question-6: Discuss the general method to solve a problem using the divide and conquer technique. Write down the binary search algorithm using the divide and conquer method.
- Question-7: What approach is used in greedy algorithms to solve problems? How is Knapsack problem solved using the greedy method? Illustrate using an example.
- Question-8: Define the concept of scheduling problems and describe a scheduling problem commonly encountered in an application of your choice. Explain the types of constraints that can make a scheduling problem infeasible.